

HOMEWORK 3

The homework has three parts. Each part must be done separately.

Write a program that predicts the outputs of an unknown function. The function takes multiple inputs and produces multiple outputs. The program uses a neural network. The neural network is feed forward network that has three layers and uses back propagation for training.

```
[2.3 4.5 3.1 | 2.6 5.3]
[0.5 3.6 4.2 | 1.9 3.2]
[2.4 1.6 7.1 | 3.2 1.5]
[6.2 3.9 5.5 | ? ? ]
```

The program uses appropriate number of hidden nodes, learning rate, and iterations. The program performs appropriate data conversion and normalization. The program reads the training data, validation data, and test data from three user specified files. The program writes the predicted outputs to a user specified file. The program displays the validation error on the screen.

Run the program on the given training file, validation file, and test file. Submit the predicted outputs, validation error, hidden nodes, learning rate, iterations, and random number seed. All six information must be in one file.

Write a program that classifies the loan applicants of a bank. The program uses a neural network. The applicants are classified based on the credit score, income, age, sex, and marital status. The applicants are classified into four classes namely high risk, medium risk, low risk, and undetermined. The training data is obtained from the past loans.

```
[671 89 55 male married | high]
[750 60 42 female single | low]
[810 70 60 male divorced | ?]
```

The program uses appropriate number of hidden nodes, learning rate, and iterations. The program performs appropriate data conversion and normalization. The program reads the training data, validation data, and test data from three user specified files. The program writes the classified data to a user specified file. The program displays the validation error on the screen.

Run the program on the given training file, validation file, and test file. Submit the classified data, validation error, hidden nodes, learning rate, iterations, and random number seed. All six information must be in one file.

Write a program that recognizes the images of digits. Specifically the program tells whether an image is zero, one, or two. An image is a 16×16 array of bits. The program uses a neural network. The images are classified based on their 256 bits. The images are classified into three classes namely zero, one, and two.

0 0 0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0 0 0
0 1 1 1 1 1 1 1 0 0	0 0 0 0 1 1 0 0 0 0	0 1 1 1 1 1 1 0 1 0
0 1 1 0 0 0 0 0 1 0	0 0 0 0 1 1 0 0 0 0	1 1 1 1 1 1 1 1 0 0
0 0 1 0 0 0 0 1 1 0	0 0 0 1 1 1 1 0 0 0	0 0 0 0 0 0 1 1 0 0
1 1 0 0 0 0 0 1 1 0	0 0 0 0 1 1 0 0 0 0	0 0 0 0 0 1 1 0 0 0
0 1 1 0 0 0 0 0 1 0	0 0 0 0 1 0 0 0 0 0	0 0 0 1 1 0 0 0 0 0
1 1 0 0 0 0 0 1 1 0	0 0 0 0 1 1 0 0 0 0	0 0 1 1 0 0 0 0 0 0
0 1 0 0 0 0 0 1 0 0	0 0 0 1 1 0 0 0 0 0	0 0 1 1 0 0 0 0 0 0
0 0 1 1 1 0 1 1 0 0	0 0 0 0 1 1 0 0 0 0	0 0 1 1 1 1 0 1 1 0
0 0 0 0 0 1 0 0 0 0	0 0 0 0 0 0 0 0 0 0	0 0 0 1 1 1 1 1 0 0

The program uses appropriate number of hidden nodes, learning rate, and iterations. The program performs appropriate data conversion. The program reads the training data, validation data, and test data from three user specified files. The program writes the classified data to a user specified file. The program displays the validation error on the screen.

Create a training file containing at least thirty images. Each training record contains one image and its label. Similarly create a validation file and a test file each containing at least six images. Run the program on the training file, validation file, and test file. Submit the classified data, validation error, hidden nodes, learning rate, iterations, and random number seed. All six information must be in one file. Also submit the training file, validation file, test file.

Program must be written Java language. Write comments explaining all steps. Run the program on the data provided separately. Submit the program and the results.